

Questions

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Q1 - 2024 (04 Apr Shift 2)

In a system two particles of masses $m_1 = 3 \text{ kg}$ and $m_2 = 2 \text{ kg}$ are placed at certain distance from each other. The particle of mass m_1 is moved towards the center of mass of the system through a distance 2 cm . In order to keep the center of mass of the system at the original position, the particle of mass m_2 should move towards the center of mass by the distance _____ cm.

Q2 - 2024 (06 Apr Shift 1)

Four particles A, B, C, D of mass $\frac{m}{2}, m, 2m, 4m$, have same momentum, respectively. The particle with maximum kinetic energy is :

(1) B (2) A (3) D (4) C

Q3 - 2024 (06 Apr Shift 2)

When kinetic energy of a body becomes 36 times of its original value, the percentage increase in the momentum of the body will be:

(1) 6%

(2) 600%

(3) 60%

(4) 500%

Q4 - 2024 (08 Apr Shift 1)

Three bodies A, B and C have equal kinetic energies and their masses are 400 g , 1.2 kg and 1.6 kg respectively. The ratio of their linear momenta is :

(1) $\sqrt{2}; \sqrt{3}; 1$ (2) $1 : \sqrt{3} : 2$

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(3) $1 : \sqrt{3} : \sqrt{2}$

(4) $\sqrt{3} : \sqrt{2} : 1$

Q5 - 2024 (08 Apr Shift 1)

A player caught a cricket ball of mass 150 g moving at a speed of 20 m/s. If the catching process is completed in 0.1 s, the magnitude of force exerted by the ball on the hand of the player is:

(1) 3 N

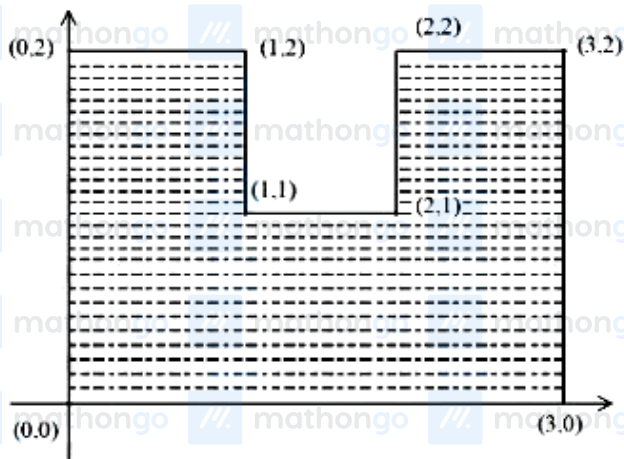
(2) 300 N

(3) 150 N

(4) 30 N

Q6 - 2024 (08 Apr Shift 1)

A uniform thin metal plate of mass 10 kg with dimensions is shown. The ratio of x and y coordinates of center of mass of plate in $\frac{n}{9}$. The value of n is _____



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Answer Key

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Q1 (3) mathongo Q2 (2) mathongo Q3 (4) mathongo Q4 (2) mathongo

Q5 (4) mathongo Q6 (15) mathongo mathongo mathongo mathongo

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Solutions

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Q1

$$\Delta X_{\text{C.O.M.}} = \frac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1 + m_2}$$

$$\Rightarrow 0 = \frac{3 \times 2 + 2(-x)}{3 + 2}$$

$$\Rightarrow x = 3 \text{ cm}$$

Q2

$$\text{KE} = \frac{p^2}{2m}$$

Same momentum, so less mass means more KE.

So $\frac{m}{2}$ will have max. KE.

Q3

$$\text{Kinetic energy } (K) = \frac{P^2}{2m}$$

$$\Rightarrow P = \sqrt{2mK}$$

$$\text{If } K_f = 36K_i$$

$$\text{So, } P_f = 6P_i$$

$$\% \text{ increase in momentum} = \frac{P_f - P_i}{P_i} \times 100\%$$

$$= \frac{6P_i - P_i}{P_i} \times 100\%$$

$$= 500\%$$

Q4

$$\text{KE} = \frac{P^2}{2m}$$

$$P \propto \sqrt{m}$$

$$\text{Hence, } P_A : P_B : P_C$$

$$= \sqrt{400} : \sqrt{1200} : \sqrt{1600} = 1 : \sqrt{3} : 2$$

Q5

$$F = \frac{\Delta P}{\Delta t} = \frac{mv - 0}{0.1}$$

$$= \frac{150 \times 10^{-3} \times 20}{0.1} = 30 \text{ N}$$

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Solutions

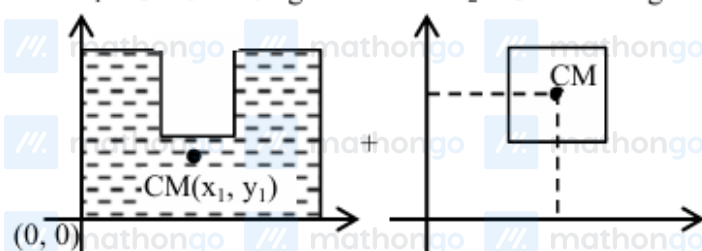
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Q6

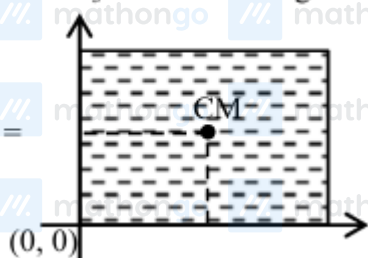
$$m_1 = \sigma \times 5 = 10\text{Kg}$$

$$m_1 = \sigma \times 5 = 10\text{Kg}$$

$$m_2 = \sigma \times 1 = 2\text{Kg}$$



$$m_3 = \sigma \times 6 = 12\text{Kg}$$



$$\Rightarrow m_1 x_1 + m_2 x_2 = m_3 x_3$$

$$10x_1 + 2(1.5) = 12(1.5) \Rightarrow x_1 = 1.5 \text{ cm}$$

$$\Rightarrow m_1 y_1 + m_2 y_2 = m_3 y_3$$

$$10y_1 + 2(1.5) = 12 \times 1 \Rightarrow y_1 = 0.9 \text{ cm}$$

$$\frac{x_1}{y_1} = \frac{1.5}{0.9} = \frac{15}{9}$$

$$n = 15$$

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